

Answer Key — Core Mathematics Grade 10

Sub-Strand 2.4: Trigonometry 1 • Lesson 2: Defining Sine, Cosine, and Tangent — Naming the Ratios That Unlock the Mast

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
C	B	A	C	D	D	B	C

USE AFTER: PREDICT PHASE — Predict: which side-ratio connects the 50° angle to the mast's height (Nakuru mast, 40 m from the base)

Q1. A student stands 40 m from the base of a mast and measures the angle from the ground up to the top as 50° . The mast's height is the side opposite the 50° angle, and the 40 m is the side adjacent to it. Which ratio links the angle to the height?

- A. $\sin 50^\circ = \text{opposite} \div \text{hypotenuse}$
- B. $\cos 50^\circ = \text{adjacent} \div \text{hypotenuse}$
- C. $\tan 50^\circ = \text{opposite} \div \text{adjacent}$**
- D. $\tan 50^\circ = \text{adjacent} \div \text{opposite}$

Correct answer: C. The height (opposite) and the 40 m distance (adjacent) are both involved and the hypotenuse is not, so the tangent ratio, $\text{opposite} \div \text{adjacent}$, is the one that links them.

USE AFTER: OBSERVE PHASE — Observe: pairs label O, A, H for each acute angle in their two triangles, noticing that labels change when the reference angle switches

Q2. In a right-angled triangle PQR, the right angle is at Q and PR is the longest side. Looking from angle P, a learner labels side QR as 'opposite'. She then switches to look from angle R. What is side QR called now?

- A. Still opposite, because the side does not move
- B. Adjacent, because it now lies next to the reference angle R**
- C. It has no label, because it is only labelled from angle P
- D. Hypotenuse, because it is a side of the triangle

Correct answer: B. Opposite and adjacent depend on the reference angle. From angle R, side QR touches the angle and is not the hypotenuse, so it is adjacent.

USE AFTER: OBSERVE PHASE — Observe: learners open the sine table and compare table values with their computed ratios

Q3. Two learners draw a right-angled triangle with an angle of about 35° . They measure the opposite side as 5.8 cm and the hypotenuse as 10.0 cm, so their $\sin 35^\circ$ is 0.58. The mathematical table gives $\sin 35^\circ = 0.574$. What best explains the small difference?

- A. Small errors in drawing and measuring the sides; the table value is the reliable one**
- B. The table is wrong because it does not match their triangle
- C. Their triangle is too small, so the sine ratio changes with size
- D. They should have used $\text{adjacent} \div \text{hypotenuse}$ for sine

Correct answer: A. Sine depends only on the angle, not the triangle's size. The small gap comes from measurement error with ruler and protractor, while the table value is pre-computed and accurate.

USE AFTER: EXPLAIN PHASE — Explain: SOH-CAH-TOA definitions and choosing the correct ratio

Q4. A 10 m ladder leans against a wall and makes an angle of 60° with the ground. The ladder is the hypotenuse, and the height it reaches up the wall is the side opposite the 60° angle. Which ratio should be used to find that height?

- A. $\tan 60^\circ = \text{adjacent} \div \text{opposite}$
- B. $\tan 60^\circ = \text{opposite} \div \text{adjacent}$
- C. $\sin 60^\circ = \text{opposite} \div \text{hypotenuse}$**
- D. $\cos 60^\circ = \text{adjacent} \div \text{hypotenuse}$

Correct answer: C. The known side is the hypotenuse and the wanted side is the opposite, so SOH (sine = opposite $\div$ hypotenuse) is the ratio to use. Tangent would need the adjacent side, which is not given.

USE AFTER: EXPLAIN PHASE — Explain: guided mast example, height = $40 \times \tan 50^\circ$

Q5. A student stands 40 m from the base of a mast and measures the angle of elevation to the top as 50° . From tables, $\tan 50^\circ \approx 1.192$. What is the height of the mast, to 1 decimal place?

- A. 25.7 m
- B. 30.6 m
- C. 33.6 m
- D. 47.7 m**

Correct answer: D. $\tan 50^\circ = \text{height} \div 40$, so height = $40 \times \tan 50^\circ = 40 \times 1.192 \approx 47.7$ m. Dividing gives 33.6 m, and using sin or cos gives the other wrong values.

USE AFTER: EXPLAIN PHASE — Explain: independent practice (a), the Kisumu tree problem

Q6. A surveyor in Kisumu stands 25 m from the base of a tree and measures the angle of elevation to the top as 38° . Using $\tan 38^\circ \approx 0.781$, what is the height of the tree, to 1 decimal place?

- A. 15.4 m
- B. 40.6 m
- C. 32.0 m
- D. 19.5 m**

Correct answer: D. $\tan 38^\circ = \text{height} \div 25$, so height = $25 \times 0.781 \approx 19.5$ m. Dividing 25 by 0.781 gives 32.0 m, which is a common error.

USE AFTER: EXPLAIN PHASE — Explain: independent practice (b), back-calculating θ from $\sin \theta = 0.643$

Q7. You know that $\sin \theta = 0.643$ for an acute angle θ and want to find θ using mathematical tables. What should you do?

- A. Divide 0.643 by 90 to get the angle
- B. Find 0.643 in the sine table and read the angle that goes with it**
- C. Find 0.643 in the cosine table and read the angle that goes with it
- D. Multiply 0.643 by 100 to get the angle in degrees

Correct answer: B. Tables list each angle with its sine value, so you can work backwards by finding the given ratio in the sine column and reading the angle. Here it gives about 40° .

USE AFTER: MODEL BUILDING PHASE — Model: adding $\sin \theta$, $\cos \theta$ and $\tan \theta$ formulas to the Lesson 1 triangle model

Q8. You are adding ratio formulas to your right-angled triangle model, which has sides labelled opposite (O), adjacent (A) and hypotenuse (H) relative to angle θ . Which formula belongs next to the adjacent side and the hypotenuse?

- A. $\tan \theta = A \div H$
- B. $\cos \theta = O \div H$
- C. $\cos \theta = A \div H$**
- D. $\sin \theta = A \div H$

Correct answer: C. Cosine is adjacent $\div$ hypotenuse (CAH), so $\cos \theta = A \div H$ goes with the adjacent side and the hypotenuse.

